

Matryoshka doll

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Matryoshka dolls are a set of wooden dolls of decreasing size placed one inside another. What's the final one?

Hints

- Wait, you can hide files inside files? But how do you find them?
 - Make sure to submit the flag as picoCTF{XXXXXX}
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Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences from binary files
 - `grep` – Filter output for specific patterns
 - `exiftool` – Read image metadata
 - `binwalk` – Detect and extract embedded files (supports recursive extraction with `-Me`)
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Complete Solution

Step 1: Download and Prepare

Download the file named `dolls.jpg` from the challenge page.

The name suggests a **Matryoshka doll** (Russian nesting doll) – a hint that there may be multiple layers of files embedded within each other.





Step 2: Basic File Inspection

First, verify the file type and general information:

```
file dolls.jpg
```

Expected output:

```
dolls.jpg: PNG image data, 594 x 1104, 8-bit/color RGBA, non-interlaced
```

Interesting observation: The file extension is `.jpg`, but the actual file type is **PNG**! This is the first clue that something is unusual.

Step 3: Search for Visible Text

Let's check if the flag appears as plain text:

```
strings dolls.jpg | grep -i "picoCTF"
```

Output: (no results)

The flag is not directly visible.

Step 4: Examine Metadata

Use `exiftool` to inspect metadata for any clues:

```
exiftool dolls.jpg
```

Output:

```
ExifTool Version Number      : 13.50
File Name                    : dolls.jpg
```

```
Directory          : .
File Size          : 652 kB
File Modification Date/Time : 2026:05:16 02:05:55+02:00
File Access Date/Time   : 2026:05:16 02:06:14+02:00
File Inode Change Date/Time : 2026:05:16 02:06:08+02:00
File Permissions      : -rw-rw-rw-
File Type           : PNG
File Type Extension   : png
MIME Type           : image/png
Image Width          : 594
Image Height         : 1104
Bit Depth            : 8
Color Type           : RGB with Alpha
Compression          : Deflate/Inflate
Filter               : Adaptive
Interlace            : Noninterlaced
Profile Name         : ICC Profile
Profile CMM Type     : Apple Computer Inc.
...
```

Nothing suspicious in the metadata, but we already know the file is actually a PNG.

Step 5: Scan with Binwalk

Since the challenge is named “Matryoshka doll” (nesting dolls), there are likely **embedded files** inside. Use `binwalk` to detect them:

```
binwalk dolls.jpg
```

Output:

DECIMAL	HEXADECIMAL	DESCRIPTION

0	0x0	PNG image, 594 x 1104, 8-bit/color RGBA, non-interlaced
3226	0xC9A	TIFF image data, big-endian, offset of first image
directory: 8		
272492	0x4286C	Zip archive data, at least v2.0 to extract, compressed
size: 378933, uncompressed size: 383920, name: base_images/2_c.jpg		
651591	0x9F147	End of Zip archive, footer length: 22

Key finding: A **ZIP archive** is embedded in the PNG file, starting at offset `272492` . This confirms the nesting doll pattern!

Step 6: Extract Embedded Files Recursively

Use `binwalk` with the `-Me` options to extract all embedded files recursively. The `-M` option enables recursive extraction, and `-e` extracts the files:

```
binwalk dolls.jpg -Me --run-as=root
```

Output (abbreviated):

```
Scan Time:      2026-05-16 12:22:21
Target File:    /home/.../ctf/picoctf/Forensics/Matryoshka doll/dolls.jpg
MD5 Checksum:  563f82d420f407f582d6cce327617ea2
Signatures:    436

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
272492      0x4286C        Zip archive data, at least v2.0 to extract, compressed
size: 378933, uncompressed size: 383920, name: base_images/2_c.jpg

WARNING: One or more files failed to extract: either no utility was found or it's
unimplemented

Scan Time:      2026-05-16 12:22:21
Target File:    /home/.../ctf/picoctf/Forensics/Matryoshka
doll/_dolls.jpg.extracted/base_images/2_c.jpg
MD5 Checksum:  e3fa93b59d62b25117c563f028cdf8ed
Signatures:    436

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
187707      0x2DD3B        Zip archive data, at least v2.0 to extract, compressed
size: 196025, uncompressed size: 201427, name: base_images/3_c.jpg

WARNING: One or more files failed to extract: either no utility was found or it's
unimplemented

Scan Time:      2026-05-16 12:22:21
Target File:    /home/.../ctf/picoctf/Forensics/Matryoshka
doll/_dolls.jpg.extracted/base_images/_2_c.jpg.extracted/base_images/3_c.jpg
MD5 Checksum:  fdbf99dfe0b7d1597c3eb005cbefccab
Signatures:    436
```

DECIMAL	HEXADECIMAL	DESCRIPTION

123606	0x1E2D6	Zip archive data, at least v2.0 to extract, compressed size: 77633, uncompressed size: 79786, name: base_images/4_c.jpg

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

```

Scan Time:      2026-05-16 12:22:21
Target File:    /home/.../ctf/picoctf/Forensics/Matryoshka
doll/_dolls.jpg.extracted/base_images/_2_c.jpg.extracted/base_images/_3_c.jpg.extrac
d/base_images/4_c.jpg
MD5 Checksum:   dbc103a52d4ea7741386f2b409fe0c68
Signatures:     436
  
```

DECIMAL	HEXADECIMAL	DESCRIPTION

79578	0x136DA	Zip archive data, at least v1.0 to extract, compressed size: 42, uncompressed size: 42, name: flag.txt

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

```

Scan Time:      2026-05-16 12:22:21
Target File:    /home/.../ctf/picoctf/Forensics/Matryoshka
doll/_dolls.jpg.extracted/base_images/_2_c.jpg.extracted/base_images/_3_c.jpg.extrac
d/base_images/_4_c.jpg.extracted/flag.txt
MD5 Checksum:   c8974fd9d4f96521b3fe75e817437d77
Signatures:     436
  
```

What happened:

- Level 1: `dolls.jpg` → extracted → `2_c.jpg`
- Level 2: `2_c.jpg` → extracted → `3_c.jpg`
- Level 3: `3_c.jpg` → extracted → `4_c.jpg`
- Level 4: `4_c.jpg` → extracted → `flag.txt`

The warnings are normal – `binwalk` successfully extracted the files despite the message.

Step 7: Read the Flag

Navigate to the final extracted file and read its contents:

```
cat
_dolls.jpg.extracted/base_images/_2_c.jpg.extracted/base_images/_3_c.jpg.extracted/bas
e_images/_4_c.jpg.extracted/flag.txt
```

Output:

```
picoCTF{<REDACTED>}
```

 Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, following these steps will reveal the complete flag.

 Key Concepts

Concept	Description
Matryoshka Doll	A Russian nesting doll where each doll contains a smaller one inside – metaphor for layered files.
File Extension Mismatch	The file had a <code>.jpg</code> extension but was actually a PNG, indicating possible tampering.
Embedded Files	Files can be hidden inside other files using tools like <code>binwalk</code> .
Recursive Extraction	Embedded files may themselves contain more embedded files, requiring multiple extraction passes.
Binwalk	A tool for detecting and extracting embedded files from within other files.

Concept	Description
PNG Metadata	The ICC Profile metadata revealed the file was created on macOS, but that wasn't directly useful.

Pro Tips

1. **Don't trust file extensions** – Always use `file` to verify the actual file type.
2. **The name is a hint** – “Matryoshka doll” suggests nested/layered files. Always look for embedded content.
3. **Use `binwalk -Me` for deep extraction** – The `-M` (recursive) and `-e` (extract) options will automatically extract nested files.
4. **Warnings can be ignored** – `binwalk` often prints warnings about missing utilities, but the extraction still works.
5. **Check the extracted directory** – After running `binwalk`, explore the `_dolls.jpg.extracted` directory to find all extracted content.
6. **PNG files can contain ZIP archives** – PNG chunks can store arbitrary data. The ZIP was likely embedded in a custom PNG chunk.